

Digital Communications

Signal - Space Analysis

Gram – Schmidt orthogonalization
Geometric Representation of Signals
Coherent Detection of Signals in Noise
Correlation Receiver
Probability of Error



Signal - Space Analysis

- **Gram – Schmidt orthogonalization**
 - The Gram–Schmidt Process is an algorithm to obtain an orthogonal (or orthonormal) base from any base
 - The Gram–Schmidt process receives a finite set, linearly independent of vectors $\mathbf{B} = \{v_1, \dots, v_n\}$ and returns an orthogonal set $\mathbf{B}' = \{w_1, \dots, w_n\}$ that generates the same initial subspace S



Signal - Space Analysis

- **Gram – Schmidt orthogonalization**
 - The projection operator is defined by:

$$\text{proj}_{\mathbf{u}} (\mathbf{v}) = \frac{\langle \mathbf{v}, \mathbf{u} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u}$$

- This operator projects the vector $\mathbf{v}$ orthogonally onto the line generated by vector $\mathbf{u}$



Signal - Space Analysis

- **Gram – Schmidt orthogonalization**

$$\mathbf{u}_1 = \mathbf{v}_1,$$

$$\mathbf{e}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|}$$

$$\mathbf{u}_2 = \mathbf{v}_2 - \text{proj}_{\mathbf{u}_1}(\mathbf{v}_2),$$

$$\mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}$$

$$\mathbf{u}_3 = \mathbf{v}_3 - \text{proj}_{\mathbf{u}_1}(\mathbf{v}_3) - \text{proj}_{\mathbf{u}_2}(\mathbf{v}_3),$$

$$\mathbf{e}_3 = \frac{\mathbf{u}_3}{\|\mathbf{u}_3\|}$$

$$\mathbf{u}_4 = \mathbf{v}_4 - \text{proj}_{\mathbf{u}_1}(\mathbf{v}_4) - \text{proj}_{\mathbf{u}_2}(\mathbf{v}_4) - \text{proj}_{\mathbf{u}_3}(\mathbf{v}_4),$$

$$\mathbf{e}_4 = \frac{\mathbf{u}_4}{\|\mathbf{u}_4\|}$$

$$\vdots$$
$$\vdots$$

$$\mathbf{u}_k = \mathbf{v}_k - \sum_{j=1}^{k-1} \text{proj}_{\mathbf{u}_j}(\mathbf{v}_k),$$

$$\mathbf{e}_k = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|}$$



Signal - Space Analysis

- **Gram – Schmidt orthogonalization**

$$w_1 = v_1$$

$$w_2 = v_2 - \frac{\langle v_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$$

$$w_3 = v_3 - \frac{\langle v_3, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_3, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2$$

$$\vdots$$

$$w_i = v_i - \frac{\langle v_i, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1 - \frac{\langle v_i, w_2 \rangle}{\langle w_2, w_2 \rangle} w_2 - \dots - \frac{\langle v_i, w_{i-1} \rangle}{\langle w_{i-1}, w_{i-1} \rangle} w_{i-1}$$



Signal - Space Analysis

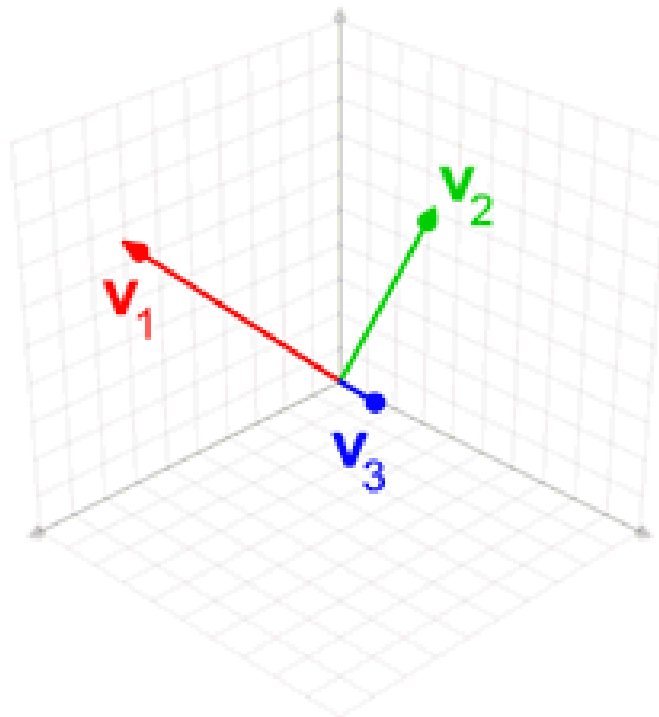
- **Gram – Schmidt orthogonalization**

- If $\mathbf{B}' = \{w_1, \dots, w_n\}$ is an orthogonal base of a space with an inner product V ,

$$\tilde{\mathbf{B}} = \left\{ \frac{w_1}{\|w_1\|}, \frac{w_2}{\|w_2\|}, \dots, \frac{w_n}{\|w_n\|} \right\}$$

- is an orthonormal base of the same subspace
- The Gram-Schmidt Theorem guarantees the existence of an infinity of orthonormal bases

Signal - Space Analysis



Signal - Space Analysis

- **Gram – Schmidt orthogonalization (Example)**

- Consider the vectors:

$$\vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{e} \quad \vec{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

- The space generated by $\{\mathbf{V}_1, \mathbf{V}_2$ and $\mathbf{V}_3\}$ is all space $\mathbf{R}^3$, as we have three linearly independent vectors in a three-dimensional space:

- Check if they are linearly independent
 - Check if they are orthogonal



Signal - Space Analysis

- **Gram – Schmidt orthogonalization (Example)**

$$\vec{u}_1 = \vec{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\vec{u}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1 \\ -1/2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$\vec{u}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{2-1}{1+4+1} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{6} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

Signal - Space Analysis

- **Gram – Schmidt orthogonalization (Example)**
 - We conclude that the set of vectors:

$$\left\{ \vec{u}_1, \vec{u}_2, \vec{u}_3 \right\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1/2 \\ 1 \\ -1/2 \end{bmatrix}, \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} \right\}$$

- is orthogonal and generates the same space as:

$$\left\{ \vec{v}_1, \vec{v}_2, \vec{v}_3 \right\}$$

Signal - Space Analysis

- **Gram – Schmidt orthogonalization (Example)**
 - We also observed that, by normalizing the vectors, we can obtain an orthonormal basis:

$$\|\vec{u}_1\| = \sqrt{1 + 0 + 1} = \sqrt{2}$$

$$\|\vec{u}_2\| = \sqrt{1 + 4 + 1} = \sqrt{6}$$

$$\|\vec{u}_3\| = \sqrt{1 + 1 + 1} = \sqrt{3}$$

$$\frac{\vec{u}_1}{\|\vec{u}_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$\frac{\vec{u}_2}{\|\vec{u}_2\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$$

$$\frac{\vec{u}_3}{\|\vec{u}_3\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$



Signal - Space Analysis

- **Geometric representation of signals**

- An N -dimensional orthogonal space is characterized by a set of N linearly independent functions, the so-called basis functions, $\{\psi_j(t)\}$
- Any arbitrary function, $s_i(t)$, in this space can be generated through a linear combination of the basis functions:

$$s_i(t) = \sum_{j=1}^N s_{ij} \psi_j(t)$$



Signal - Space Analysis

- **Geometric representation of signals**

- The coefficients are given by:

$$s_{ij} = \frac{1}{K_j} \int_0^T s_i(t) \psi_j(t) dt \quad K_j - \text{constant}$$

- The basis functions must satisfy the condition:

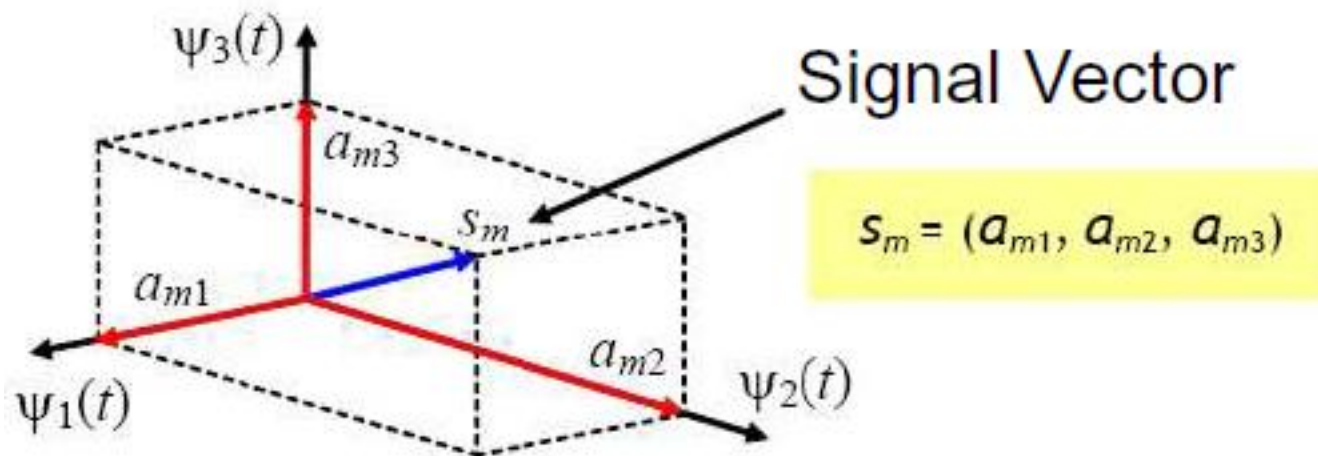
$$\int_0^T \psi_j(t) \psi_k(t) dt = \begin{cases} K_j & j = k \\ 0 & j \neq k \end{cases} \quad \begin{matrix} 0 \leq t \leq T \\ j, k = 1, 2, \dots, N \end{matrix}$$

- Orthonormal space $\Rightarrow K_j = 1$
 - Orthogonal space $\Rightarrow K_j \neq 0$



Signal - Space Analysis

- **Geometric representation of signals**
 - Example of orthogonal space with $N = 3$



Signal - Space Analysis

- **Geometric representation of signals**

- The signal $s_i(t)$ is represented by the vector:
- $s_i = [s_1, s_2, \dots, s_N]^T$
- The square of the length, or norm, of the vector is defined as:

$$\|s_i\|^2 = s_i^T s_i = \sum_{j=1}^N s_{ij}^2 \quad i = 1, 2, \dots, M$$

- It is the inner product of the vector with itself



Signal - Space Analysis

- **Geometric representation of signals**
 - Correspondence between the energy of a signal and its vector representation:

$$E_i = \int_0^T s_i^2(t) dt$$

$$E_i = \int_0^T \left[\sum_{j=1}^N s_{ij} \psi_j(t) \right] \left[\sum_{k=1}^N s_{ik} \psi_k(t) \right] dt = \sum_{j=1}^N \sum_{k=1}^N s_{ij} s_{ik} \int_0^T \psi_j(t) \psi_k(t) dt$$

Signal - Space Analysis

- **Geometric representation of signals**

- This expression is simplified a lot since:

$$\int_0^T \psi_j(t) \psi_k(t) dt = \delta_{jk}$$

- and the energy is given by:

$$E_i = \sum_{j=1}^N s_{ij}^2 = \|\mathbf{s}_i\|^2$$

- The energy of a signal is equal to the square norm of the vector that represents it



Signal - Space Analysis

- **Geometric representation of signals**

- Euclidean distance between two points:

$$d_{ik} = \|\mathbf{s}_i - \mathbf{s}_k\|$$

$$d_{ik}^2 = \|\mathbf{s}_i - \mathbf{s}_k\|^2 = \sum_{j=1}^N (s_{ij} - s_{kj})^2 = \int_0^T [s_i(t) - s_k(t)]^2 dt = E_d$$

- The energy of the difference of two signals is equal to the squared distance between the corresponding points in space



Signal - Space Analysis

- **Geometric representation of signals**
 - Correlation between two signals is their inner product:

$$\int_0^T s_i(t) s_k(t) dt = \mathbf{s}_i^T \mathbf{s}_k$$

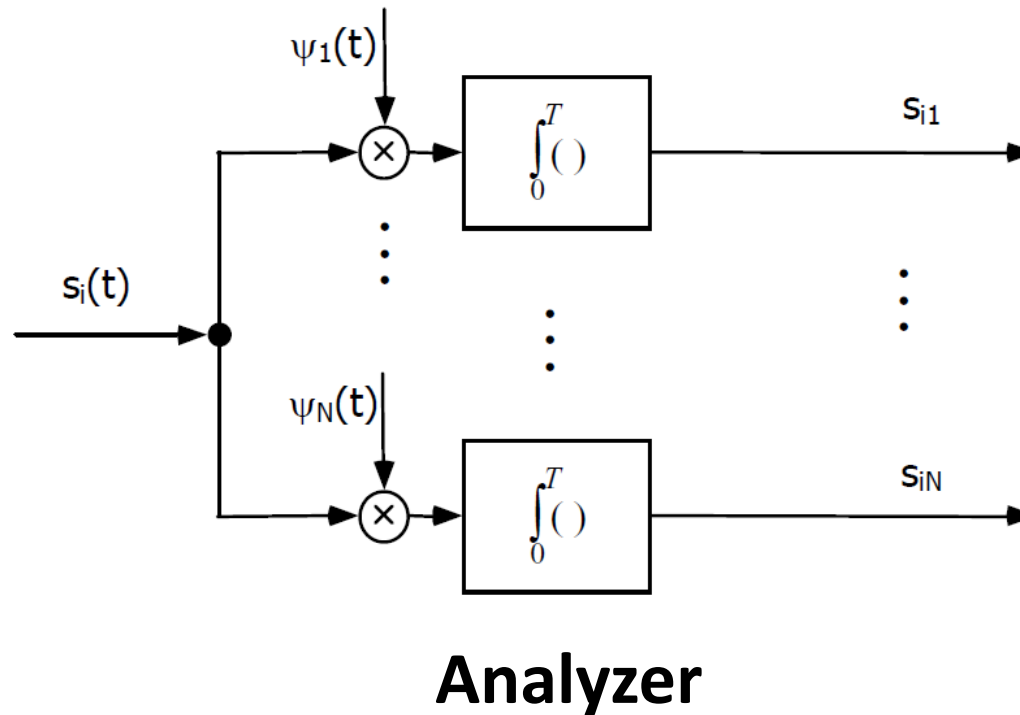
- Angle between two signal vectors:

$$\cos \theta_{ik} = \frac{\mathbf{s}_i^T \mathbf{s}_k}{\|\mathbf{s}_i\| \|\mathbf{s}_k\|}$$



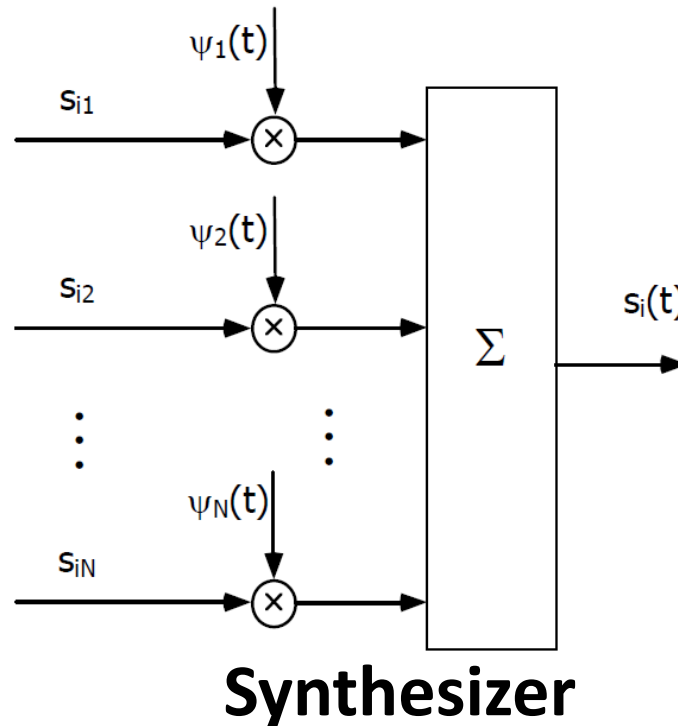
Signal - Space Analysis

- **Geometric representation of signals**
 - How to get the elements of vector s_i from $s_i(t)$?



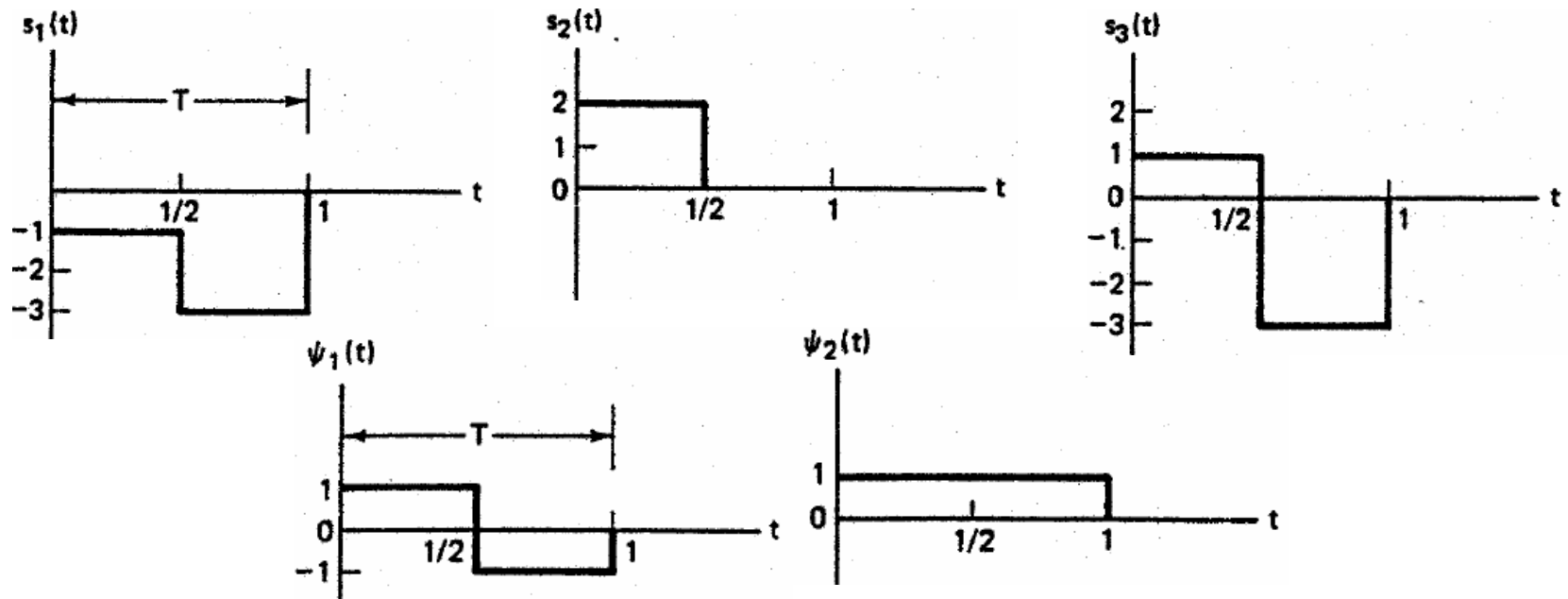
Signal - Space Analysis

- **Geometric representation of signals**
 - How to get $s_i(t)$ from the elements of vector s_i ?



Signal - Space Analysis

- **Geometric representation of signals**
 - Example of a set of arbitrary signals defined as function of orthogonal signals



Signal - Space Analysis

- **Geometric representation of signals**

- The set $\{s_i(t)\}$ is not orthogonal because (example for $s_1(t)$ and $s_2(t)$):

$$\int_0^T s_1(t)s_2(t)dt = \int_0^{T/2} s_1(t)s_2(t)dt + \int_{T/2}^T s_1(t)s_2(t)dt = -T$$

- to be orthogonal it would have to be:

$$\int_0^T s_i(t)s_j(t)dt = 0 \quad i \neq j$$



Signal - Space Analysis

- **Geometric representation of signals**

- The set $\{\psi_i(t)\}$ is orthogonal because:

$$\int_0^T \psi_i(t) \psi_j(t) dt = 0 \quad i \neq j$$

- We can express the non-orthogonal set $\{s_i(t)\}; i = 1, 2, 3$

- as function of the basis waveforms $\{\psi_i(t)\}; j = 1, 2$

$$\begin{cases} s_1(t) = \psi_1(t) - 2\psi_2(t) \\ s_2(t) = \psi_1(t) + \psi_2(t) \\ s_3(t) = 2\psi_1(t) - \psi_2(t) \end{cases} \quad (k_j = T)$$



Signal - Space Analysis

- **Geometric representation of signals**
 - How to obtain the basis functions (Gram-Schmidt orthogonalization)?
 - From M energy signals $s_i(t)$, represented by $\mathbf{s}_i$ vectors, we can obtain $N \leq M$ basis functions $\psi_i(t)$ proceeding as follows:
 1. One of the signals is arbitrarily chosen. Let $s_1(t)$. The first basis function is defined like this:

$$\psi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} \quad E_1 = \int_0^T s_1^2(t) dt$$



Signal - Space Analysis

- **Geometric representation of signals**

2. The first element of s_2 is by definition:

$$s_{21} = \int_0^T s_2(t) \psi_1(t) dt \rightarrow \text{Projection of } s_2 \text{ over the first basis function}$$

3. An intermediate function $g_2(t)$ is defined:

$$g_2(t) = s_2(t) - s_{21} \psi_1(t)$$

4. The second basis function is defined as function of $g_2(t)$ and its energy:

$$\psi_2(t) = \frac{g_2(t)}{\sqrt{\int_0^T g_2^2(t) dt}}$$



Signal - Space Analysis

- **Geometric representation of signals**

- Replacing $g_2(t)$:

$$\psi_2(t) = \frac{s_2(t) - s_{21}\psi_1(t)}{\sqrt{\int_0^T [s_2(t) - s_{21}\psi_1(t)]^2 dt}} = \frac{s_2(t) - s_{21}\psi_1(t)}{\sqrt{E_2 - s_{21}^2}}$$

5. Then:

$$g_i(t) = s_i(t) - \sum_{j=1}^{i-1} s_{ij}\psi_j(t)$$
$$\psi_i(t) = \frac{g_i(t)}{\sqrt{\int_0^T g_i^2(t) dt}} \quad i = 1, 2, \dots, N$$

Signal - Space Analysis

- **Geometric representation of signals**
 - Gram-Schmidt orthogonalization – (Notes)

1. This definition implies that:

$$s_{11} = \sqrt{E_1}$$

2. At this point we already have s_{21} . The other components of s_2 are included in the intermediate function $g_2(t)$. Note that $g_2(t)$ is orthogonal to $\psi_1(t)$ in the range 0 to T :

$$\int_0^T g_2(t)\psi_1(t)dt = \int_0^T s_2(t)\psi_1(t)dt - \int_0^T s_{21}\psi_1^2(t)dt = 0$$



Signal - Space Analysis

- **Geometric representation of signals**
 - Gram-Schmidt orthogonalization – (Notes)

3. It can be confirmed that:

$$\int_0^T \psi_2^2(t) dt = 1 \quad \text{and} \quad \int_0^T \psi_1(t) \psi_2(t) dt = 0$$

4. This definition encompasses particular cases. For example:

$$i = 1: g_1(t) = s_1(t)$$

$$i = 2: g_2(t) = s_2(t) - s_{21}\psi_1(t)$$



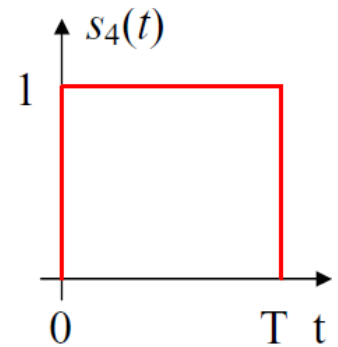
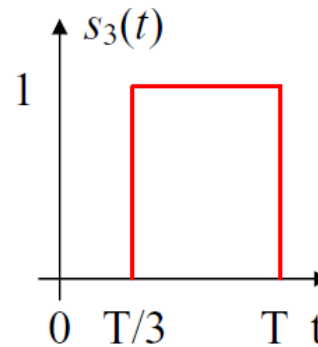
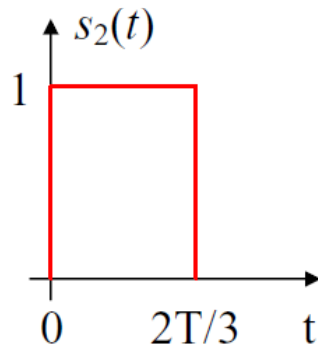
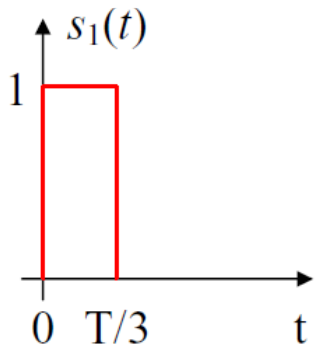
Signal - Space Analysis

- **Geometric representation of signals**
 - Gram-Schmidt orthogonalization – (Notes)
- 5. The number of basis functions, N , is less than or equal to the number of signals, M :
 - If M signals are linearly independent: $N = M$
 - If they are not linearly independent, then $N < M$ and the intermediate function $g_i(t)$ is null for $i > N$



Signal - Space Analysis

- **Geometric representation of signals**
 - Gram-Schmidt orthogonalization – An example
 - What is the orthonormal basis for this set of signals?

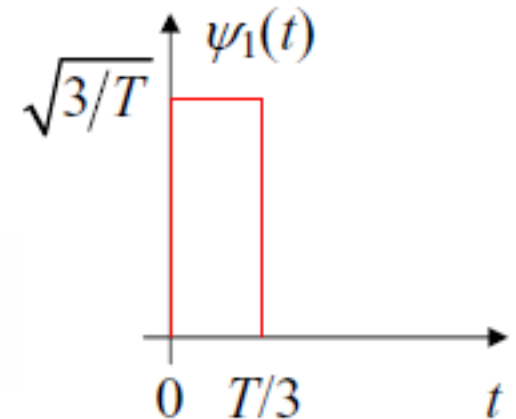


Signal - Space Analysis

- **Geometric representation of signals**
 - Gram-Schmidt orthogonalization – An example
 - 1st basis function:

$$\psi_1(t) = \frac{s_1(t)}{\sqrt{E_1}} = \begin{cases} \sqrt{3/T} & 0 \leq t \leq T/3 \\ 0 & \text{otherwise} \end{cases}$$

$$E_1 = \int_0^T s_1^2(t) dt = \int_0^{T/3} (1)^2 dt = T/3$$

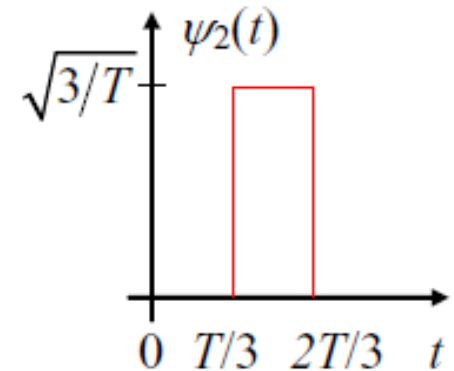


Signal - Space Analysis

- **Geometric representation of signals**
 - Gram-Schmidt orthogonalization – An example
 - 2nd basis function:
 - By definition: $s_{21} = \int_0^T s_2(t)\psi_1(t)dt = \int_0^{T/3} (1)\sqrt{3/T}dt = \sqrt{T/3}$
 - Energy of $s_2(t)$: $E_2 = \int_0^T s_2^2(t)dt = \int_0^{2T/3} (1)^2 dt = 2T/3$

$$g_2(t) = s_2(t) - s_{21}\psi_1(t) = 1 \quad T/3 \leq t \leq 2T/3$$

$$\Rightarrow \psi_2(t) = \frac{s_2(t) - s_{21}\psi_1(t)}{\sqrt{E_2 - s_{21}^2}} = \begin{cases} \sqrt{3/T} & T/3 \leq t \leq 2T/3 \\ 0 & \text{elsewhere} \end{cases}$$



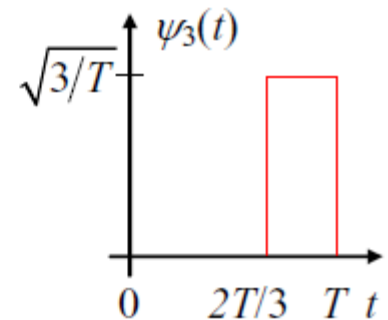
Signal - Space Analysis

- **Geometric representation of signals**
 - Gram-Schmidt orthogonalization – An example
 - 3rd basis function:

$$s_{31} = \int_0^T s_3(t) \psi_1(t) dt = 0 \quad s_{32} = \int_0^T s_3(t) \psi_2(t) dt = \sqrt{T/3}$$

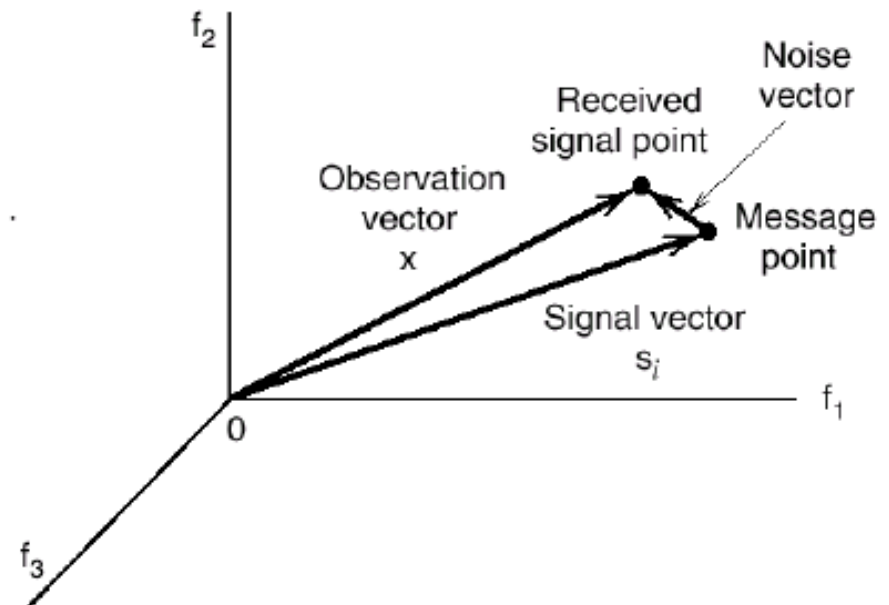
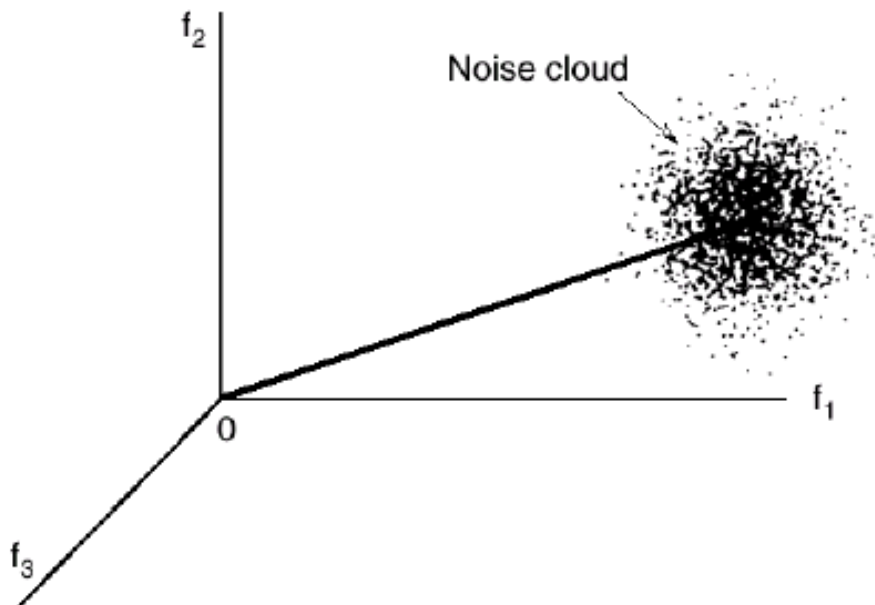
$$g_3(t) = s_3(t) - s_{31}\psi_1(t) - s_{32}\psi_2(t) = \begin{cases} 1 & 2T/3 \leq t \leq T \\ 0 & \text{otherwise} \end{cases}$$

$$\psi_3(t) = \frac{g_3(t)}{\sqrt{\int_0^T g_3^2(t) dt}} = \begin{cases} \sqrt{T/3} & 2T/3 \leq t \leq T \\ 0 & \text{otherwise} \end{cases}$$



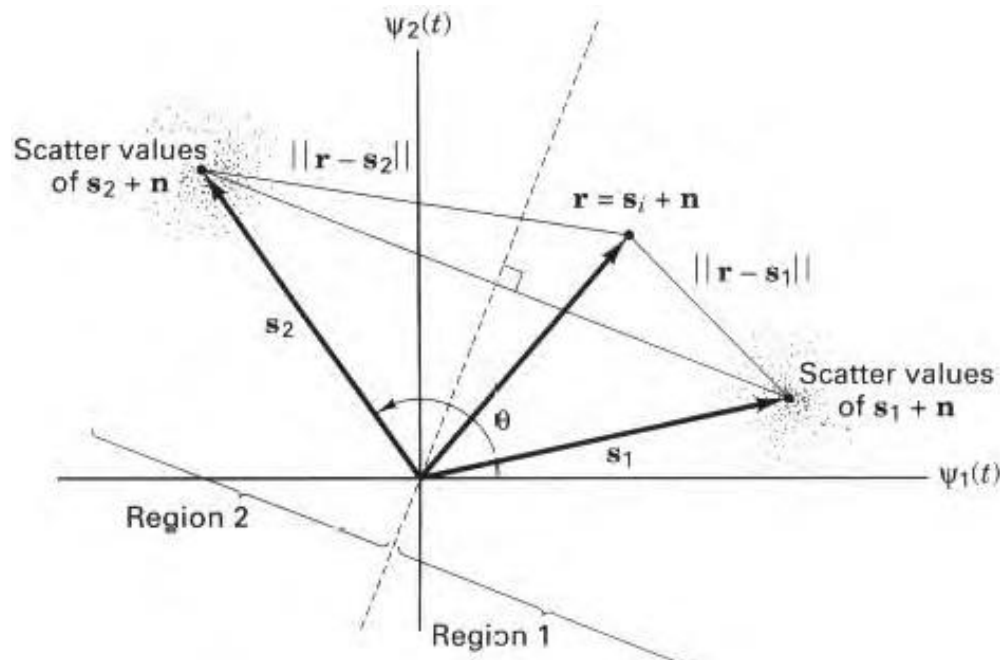
Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Decision regions – 3D



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Decision regions - 2D
 - Consider the 2-dimensional signal space



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Decision regions - 2D
 - The two-dimensional signal space is the locus of the noise-perturbed binary vectors $(s_1 + \mathbf{n})$ and $(s_2 + \mathbf{n})$
 - The noise vector, $\mathbf{n}$, is a zero-mean random vector
 - The received signal vector, $\mathbf{r}$, is a random vector with mean s_1 or s_2
 - The detector's task after receiving $\mathbf{r}$ is to decide which of the signals (s_1 or s_2) was actually transmitted

Signal - Space Analysis

- **Coherent Detection of Signals in Noise**

- Decision regions - 2D

- The method is usually to decide on the signal classification that yields the minimum expected P_E although other strategies are possible
 - For the case where M equals 2, with s_1 and s_2 being equally likely and with the noise being an AWGN process, we will see that the minimum-error decision rule is equivalent to choosing the signal class such that the distance $d(r, s_i) = \|r - s_i\|$ is minimized. where $\|x\|$ is called the norm or magnitude of vector x



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**

- Decision regions - 2D

- Let us construct decision regions in the following way:

- Draw a line connecting the tips of the prototype vectors s_1 and s_2
 - Construct the perpendicular bisector of the connecting line
 - Notice that this bisector passes through the origin of the space if s_1 and s_2 are equal in amplitude
 - For this $M = 2$ example, the constructed perpendicular bisector represents the locus of points equidistant between s_1 and s_2 ; hence the bisector describes the boundary between decision region 1 and decision region 2



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Decision regions – 3D
 - The decision rule for the detector, stated in terms of decision regions, is as follows:
 - Whenever the received signal $\mathbf{r}$ is located in region 1, choose signal s_1 ; when it is located in region 2, choose signal s_2
 - If the angle θ equals 180° then the signal set s_1 or s_2 represents BPSK



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**

- **Correlation receiver** (Introduction)

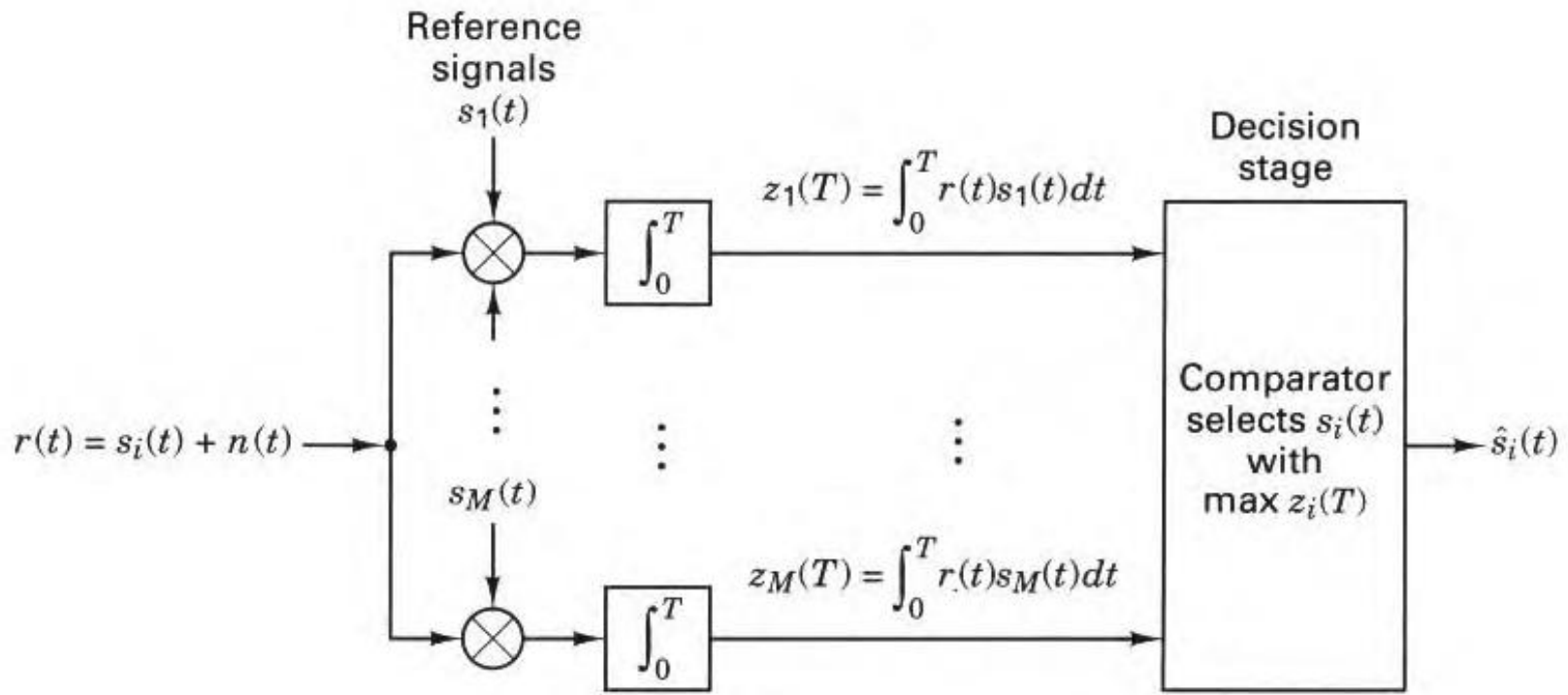
- We focus particularly on that realization of a matched filter known as a correlator
 - In addition to binary detection, we also consider the more general case of M -ary detection
 - We assume that the only performance degradation is due to AWGN. The received signal is the sum of the transmitted prototype signal plus the random noise:

$$r(t) = s_i(t) + n(t) \quad \begin{array}{l} 0 \leq t \leq T \\ i = 1, \dots, M \end{array}$$



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Correlation receiver



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**

- Correlation receiver

- A correlation receiver comprised of M correlators transforms a received waveform, $r(t)$, to a sequence of M numbers or correlator outputs, $z_i(T)$ ($i = 1, \dots, M$)
 - Each correlator output is characterized by the following product integration or correlation with the received signal:

$$z_i(T) = \int_0^T r(t)s_i(t) dt \quad i = 1, \dots, M$$

Signal - Space Analysis

- **Coherent Detection of Signals in Noise**

- Correlation receiver

- The verb "to correlate" means "to match". The correlators attempt to match the incoming received signal, $r(t)$, with each of the candidate prototype waveforms $s_i(t)$, known a priori to the receiver
 - A reasonable decision rule is to choose the waveform, $s_i(t)$, that matches best or as the largest correlation with $r(t)$
 - In other words, the decision rule is:
 - Choose the $s_i(t)$ whose index corresponds to the $\max z_i(T)$



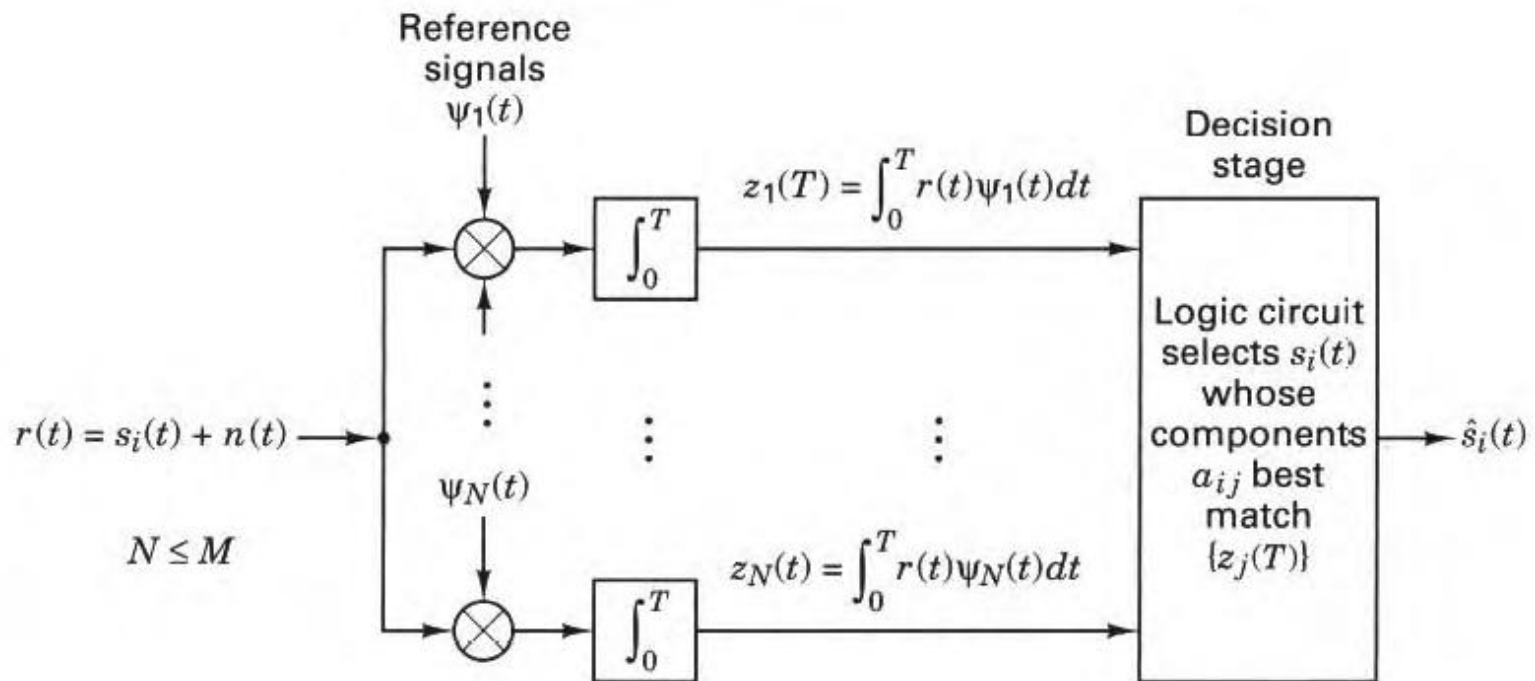
Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Correlation receiver
 - As we saw, any signal set, $\{s_i(t)\}$ ($i = 1, \dots, M$), can be expressed in terms of some set of basis functions, $\{\psi_j(t)\}$ ($j = 1, \dots, N$), where $N \leq M$
 - Then the bank of M correlators in the previous figure may be replaced with a bank of N correlators, shown in next figure, where the set of basis functions $\{\psi_j(t)\}$ form reference signals



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Correlation receiver



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Correlation receiver
 - The decision stage of this receiver consists of logic circuitry for choosing the signal $s_i(t)$
 - The choice of $s_i(t)$ is made according to the best match of the coefficients, a_{ij}
 - When the prototype waveform set, $\{s_i(t)\}$, is an orthogonal set, the receiver implementation in the two figures are identical (differing perhaps by a scale factor)



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Correlation receiver
 - However, when $\{s_i(t)\}$ is not an orthogonal set, the receiver in the last figure, using N correlators instead of M , with reference signals $\psi_j(t)$, can represent a cost-effective implementation
 - Since N is normally much smaller than M (e.g. 256-QAM has $M = 256$ while $N = 2$) this receiver is preferable

Signal - Space Analysis

- **Coherent Detection of Signals in Noise**

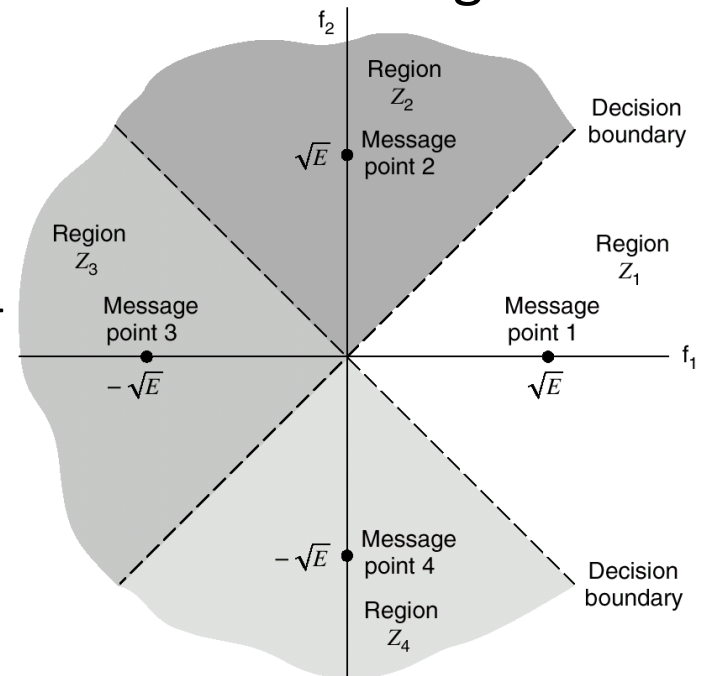
- Probability of error

- Partition of the observation space into decision regions:

- $Z_i (i = 1, 2, \dots, M)$

- Example:

- Constellation with $N = 2$ and $M = 4$

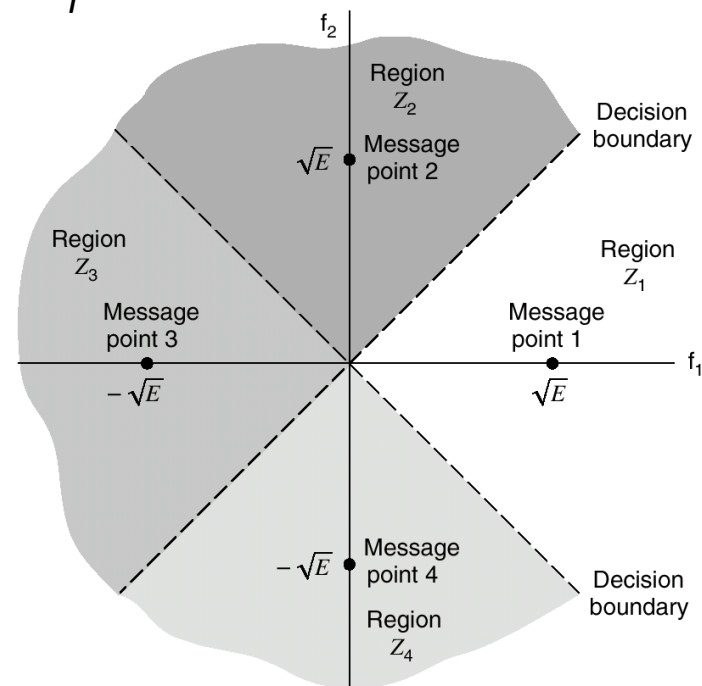


Signal - Space Analysis

- **Coherent Detection of Signals in Noise**

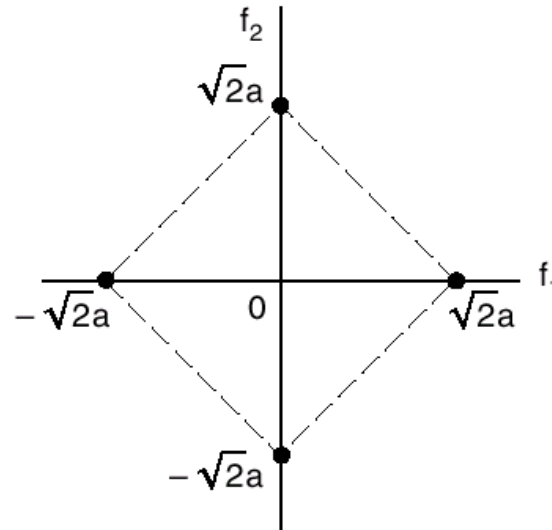
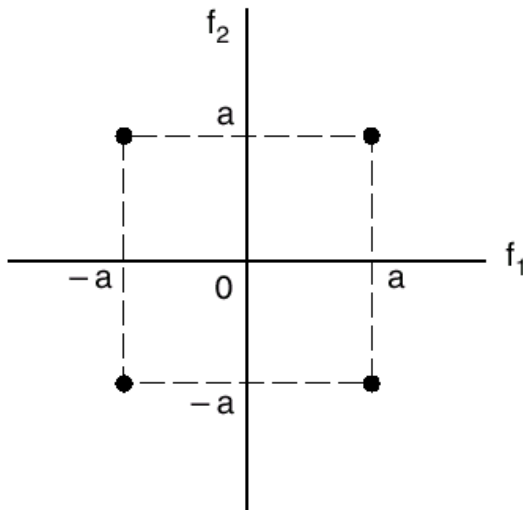
- Probability of error

- Observation vector $\mathbf{x}$ lies in Z_i if euclidean distance:
 - $\|\mathbf{x} - \mathbf{s}_k\|$
 - is minimum for $k = i$



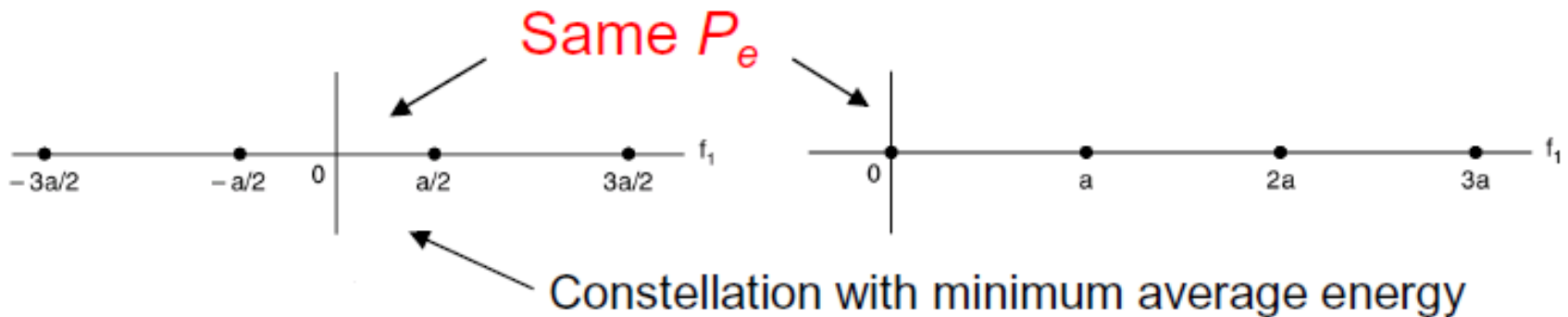
Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Probability of error
 - Rotational invariance



Signal - Space Analysis

- **Coherent Detection of Signals in Noise**
 - Probability of error
 - Translational invariance



Reading Material

- *Simon Haykin, Communication Systems, McMaster Univ, 5th Edition, 2009 (ISBN 978-047-16979-0-9)*
- *A. Bruce Carlson and Paul Crilly. Communication Systems – An Introduction to Signals and Noise, McGraw-Hill, 5th edition, 2009 (ISBN 978-007-338-040-7)*
- *Bernard Sklar and Fredric Harris, Digital Communications: Fundamentals and Applications, Pearson, 3rd Edition, 2020 (ISBN-13: 978-0134588568)*



Reading Material

